

NATIONAL TEXTILE UNIVERSITY

FAISALABAD



Title	Smart Medicine Dispenser
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SMART MEDICINE DISPENSER

INTRODUCTION

In the fast-paced world of today, patients with chronic illnesses or old age find it difficult to take their medication on time, resulting in health complications. To solve this problem, we produced a Smart Medicine Box as a digital logic design project that would assist patients in sticking to their medication regimen. This project employs an Arduino Uno R3 microcontroller to design a lowcost, easy-to-use, and straightforward device that reminds people to take their medications at the appropriate time.

The Smart Medicine Box is programmed to remind patients at scheduled times (e.g., 8:00 AM, 12:00 PM, 6:00 PM, 10:00 PM) by ringing a buzzer and illuminating an LED. Upon hearing the ring and seeing the illuminated LED, the patient pushes a button to confirm the reminder, silencing the buzzer and LED. The patient then manually takes their medication.

Apparatus:

- ✦ Arduino Uno R3
- ✦ Servo Motor
- ✦ LED
- ✦ Buzzer
- ✦ Bread Board
- ✦ Jumper Wires (Male-to-Male)
- ✦ Jumper Wires (Male-to-Female)
- ✦ Push Button
- ✦ 9 Volt Battery

Tools:

- ✦ Arduino IDE
- ✦ Tinker Cad

Source code:

□ Header Files (Libraries)

```
#include <Wire.h>
```

```
#include <LiquidCrystal_I2C.h>
```

```
#include <Servo.h>
```

- ✦ Wire.h: Enables I2C communication (used for the LCD).
- ✦ LiquidCrystal_I2C.h: Controls the I2C-based LCD.
- ✦ Servo.h: Controls the servo motor.

□ Object Definitions

LiquidCrystal_I2C lcd (0x27, 16, 2);

Servo servo;

- ✦ lcd (0x27, 16, 2): Defines an LCD with address 0x27, 16 columns, and 2 rows.
- ✦ Servo servo: Creates a servo object to control the motor.

□ Pin Assignments

const int servoPin = 5;

const int buzzerPin = 8;

const int ledPin = 7;

const int buttonPin = 4;

- ✦ Servo is connected to pin **D5**
- ✦ Buzzer to **D8**
- ✦ LED to **D7** ✦ Push button to **D4**

□ Setup Function

```
void setup() {  servo.attach(servoPin);  pinMode (buzzerPin,
OUTPUT);  pinMode (ledPin, OUTPUT);  pinMode (buttonPin,
INPUT_PULLUP); // Button connected to GND
```

```
  lcd. init();
```

```
  lcd.backlight();
```

```
lcd.setCursor(0, 0);  
lcd.print("System Starting..."); delay  
(2000); lcd.clear();  
}
```

What happens here:

- ✦ Attaches the servo to pin D5.
- ✦ Sets buzzer and LED as output.
- ✦ Sets the button as input with a pull-up resistor.
- ✦ Initializes the LCD and turns on the backlight.
- ✦ Displays “System Starting...” on LCD for 2 seconds.

▣ Main Loop

```
void loop () {
```

▣ Show Alert

```
lcd.setCursor(0, 0); lcd.print  
("Take Medicine!");  
digitalWrite (buzzerPin, HIGH);  
digitalWrite (ledPin, HIGH);
```

- ✦ Displays "**Take Medicine!**" on LCD.
- ✦ Turns ON the buzzer and LED as an alert.

▣ Wait for Button Press

```
while (digitalRead(buttonPin) == HIGH) {  
  // Waiting  
}
```

- ✦ Waits for the user to press the button.

- ✦ Button press pulls the signal LOW and exits the loop.

✧ **Stop Alert**

```
digitalWrite (buzzerPin, LOW);
```

```
digitalWrite (ledPin, LOW);
```

```
lcd. clear (); lcd. setCursor (0, 0);
```

```
lcd.print ("Dispensing..."); ✦ Turns
```

OFF the buzzer and LED.

- ✦ Displays “Dispensing...” on LCD.

✧ **Dispense Medicine (Move Servo)**

```
servo. Write (90); // Open position delay
```

```
(10000); // Wait 10 seconds servo.
```

```
Write (0); // Close position ✦ Rotates
```

servo to 90° (dispense position).

- ✦ Waits 10 seconds (time for medicine to fall).
- ✦ Returns servo to 0° (closed position).

✧ **Show Confirmation**

```
lcd. clear (); lcd.print
```

```
("Medicine Taken"); delay
```

```
(2000); lcd.clear();
```

- ✦ Shows “Medicine Taken” on LCD for 2 seconds.
- ✦ Then clears the screen.

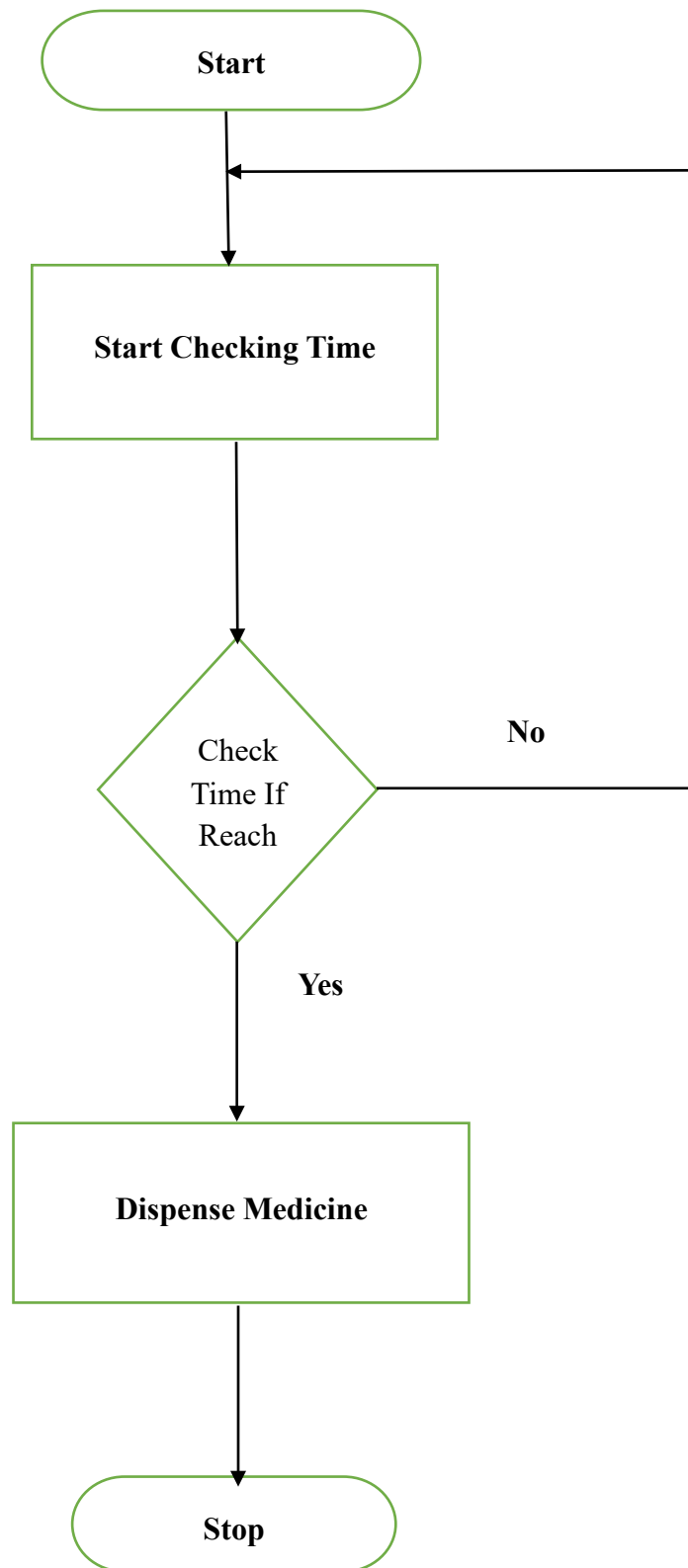
✧ **Wait Before Restart**

```
Delay (5000);
```

```
}
```

- ✦ Wait 5 seconds before repeating the loop.

Flowchart:



Project pictures:



Procedure:

✧ **System Initialization**

- ✦ LCD starts with the message: System Starting...
- ✦ Delay of 2 seconds for system startup.
- ✦ LCD is cleared after that.

✧ **Alerting the User**

- ✦ LCD displays: **"Take Medicine!"**
- ✦ **Buzzer and LED** turn ON to alert the user that it's time to take medicine.

✧ **Wait for Button Press**

- ✦ The system **waits until the user presses the push button.**
- ✦ Button is connected with INPUT_PULLUP, so it reads HIGH by default and becomes LOW when pressed.

✧ **Stop Alert**

- ✦ Once the button is pressed:
 - **Buzzer and LED are turned OFF**
 - LCD shows: **"Dispensing.."**

✧ **Dispense Medicine**

- ✧ Servo motors turns to **90 degrees** to dispense the medicine.
- ✧ It stays open for **10 seconds**.
- ✧ Then it returns to **0 degrees** (closed position).

✧ **Confirmation Message**

- ✧ LCD shows: **"Medicine Taken"**
- ✧ Stays for 2 seconds, then LCD is cleared.

✧ **Wait Before Next Cycle**

- ✧ System waits **5 seconds** before starting the loop again.

Conclusion:

When the Power is given to the Arduino it will start counting time and when the time is completed it will turn on the LED and Buzzer start ringing and motor will rotate and dispense the medicine. When the patient takes the medicine then he will also press the push button to ensure that the medicine has been taken and after pressing the push button the LED and Buzzer will stop. Then, again the countdown will start for further distribution of medicine.

THE END

